Practical 2(Option 1): Write a program to implement multimode Hadoop Cluster.

Since we are working on a single Windows 11 machine with 16GB of RAM, we physically only have one computer. To implement a true Multi-Node cluster, we would typically use multiple physical PCs. However, we can simulate this environment using your current setup by understanding the architecture and configuring the required files.

Here is exactly how a Multi-Node cluster is implemented and the configuration "code" (XML) required to make it work.

The Multi-Node Architecture

In a Multi-Node setup, Hadoop uses a Master-Worker (or Master-Slave) architecture. The roles are split across different computers.

Node Type	Role in Cluster	Hadoop Components Running
Master Node	Manages the cluster, stores metadata, and assigns tasks.	NameNode, ResourceManager
Worker Node 1	Stores actual data blocks and executes processing tasks.	DataNode, NodeManager
Worker Node 2+	Additional storage and processing power.	DataNode, NodeManager

The Configuration "Code" for Multi-Node

If you had three laptops sitting on your desk (one Master, two Workers), you would need to change the XML files you edited in Practical 1 to connect them over the network. Here are the exact configuration changes required for a Multi-Node setup.

1. The `core-site.xml` File (Applied to ALL nodes)

Instead of pointing to `localhost`, you must point every machine to the Master Node's IP address on your network.

```
<configuration>
    <property>
        <name>fs.defaultFS</name>
        <value>hdfs://192.168.1.100:9000</value>
    </property>
</configuration>
```

2. The `hdfs-site.xml` File (Applied to ALL nodes)

In a multi-node cluster, you want data redundancy. You change the replication factor from `1` to `2` or `3`, meaning every file is duplicated across multiple Worker nodes in case one crashes.

```
<configuration>
    <property>
        <name>dfs.replication</name>
```

```
        <value>2</value>
    </property>
    <property>
        <name>dfs.namenode.name.dir</name>
        <value>/C:/hadoop/data/namenode</value>
    </property>
    <property>
        <name>dfs.datanode.data.dir</name>
        <value>/C:/hadoop/data/datanode</value>
    </property>
</configuration>
```

3. The `workers` File (Applied to the MASTER node)

This is the most critical file for a Multi-Node cluster. It is a simple text file located in `C:\hadoop\etc\hadoop\workers` (in older Hadoop versions, this was called the `slaves` file). The Master node reads this file to know exactly which machines in the network are its workers. You simply list the IP addresses of the Worker nodes.

Plaintext

```
192.168.1.101
192.168.1.102
```

How to Execute and Demonstrate This for Your Practical

Method A: The Theoretical Configuration (Recommended)

Present the XML configurations above. Explain that a Multi-Node cluster is an infrastructure setup, not a programming script, and showcase how `core-site.xml`, `hdfs-site.xml`, and the `workers` file bridge the network.

Method B: The Virtualization Route If your professor strictly requires a working Multi-Node demonstration, you will need to utilize your 16GB of RAM to run virtual machines. You would install a program like **VirtualBox**, create two lightweight Ubuntu Linux virtual machines, assign them 4GB of RAM each, and install Hadoop on both of them. You would configure one VM as the Master and the other as the Worker using the exact XML configurations provided above.

Practical 2 (Option 2): Write a program to implement multimode Hadoop Cluster.

Because Windows does not natively support SSH (which Hadoop uses to automatically start Worker nodes from the Master), we will configure the network and then start the specific services manually on each machine.

Here is customized, step-by-step instructions for a 2-node Windows Hadoop cluster.

Phase 1: Pre-Flight Checklist (Do this on BOTH PCs)

Before editing any configurations, ensure these baseline rules are met on both the Master (`172.16.0.81`) and the Worker (`172.16.0.82`):

- **Identical Paths:** Ensure Java 8 and Hadoop are installed in the exact same directory paths (e.g., `C:\hadoop`) on every PC.
- **Firewall:** Windows Defender or your network firewalls will block Hadoop's internal communication. You must either temporarily disable the firewall for the local network or explicitly open Hadoop's ports (e.g., 9000 for HDFS, 8088 for YARN).
- **Clear Old Data:** If you ran Practical 1 on either of these machines, delete the `C:\hadoop\data\namenode` and `C:\hadoop\data\datanode` folders and recreate them empty. This prevents cluster ID mismatches.

Phase 2: Master Node Configuration (IP: 172.16.0.81)

Go to your Master PC, navigate to `C:\hadoop\etc\hadoop`, and edit the following files.

1. core-site.xml You must point the configuration to the Master Node's IP address on your network.

```
<configuration>
    <property>
        <name>fs.defaultFS</name>
        <value>hdfs://172.16.0.81:9000</value>
    </property>
</configuration>
```

2. hdfs-site.xml

Since you have exactly one Worker node storing data, set the replication factor to `1`.

```
<configuration>
    <property>
        <name>dfs.replication</name>
        <value>1</value>
    </property>
    <property>
        <name>dfs.namenode.name.dir</name>
        <value>/C:/hadoop/data/namenode</value>
    </property>
```

```
</configuration>
```

3. yarn-site.xml

Tell YARN where the ResourceManager (Master) is located.

```
<configuration>
    <property>
        <name>yarn.resourcemanager.hostname</name>
        <value>172.16.0.81</value>
    </property>
    <property>
        <name>yarn.nodemanager.aux-services</name>
        <value>mapreduce_shuffle</value>
    </property>
</configuration>
```

4. workers file This is a simple text file located in `C:\hadoop\etc\hadoop\workers`. The Master node reads this file to know exactly which machines in the network are its workers. You simply list the IP addresses of the Worker nodes. Delete `localhost` and add your Worker IP:

```
172.16.0.82
```

Phase 3: Worker Node Configuration (IP: 172.16.0.82)

Go to your Worker PC. You need the exact same network awareness as the Master.

1. Navigate to `C:\hadoop\etc\hadoop`.
2. Copy the exact `core-site.xml` and `yarn-site.xml` configurations you just created on the Master and paste them into the Worker's files.
3. Edit the Worker's `hdfs-site.xml` to specify its datanode directory:
XML

```
<configuration>
    <property>
        <name>dfs.replication</name>
        <value>1</value>
    </property>
    <property>
        <name>dfs.datanode.data.dir</name>
        <value>/C:/hadoop/data/datanode</value>
    </property>
</configuration>
```

Phase 4: Execution & Demonstration

Because you are on Windows without SSH, do not use the `start-dfs.cmd` or `start-all.cmd` scripts. You will start the specific roles manually on each machine.

Step 1: Format the Master (Run ONLY on Master: 172.16.0.81)

Open Command Prompt as Administrator and run:

```
hdfs namenode -format
```

Step 2: Start the Master Daemons (Run on Master: 172.16.0.81)

Open two separate Command Prompt windows as Administrator and run one command in each to keep them running:

- CMD Window 1: `hdfs namenode`
- CMD Window 2: `yarn resourcemanager`

Step 3: Start the Worker Daemons (Run on Worker: 172.16.0.82)

Open two separate Command Prompt windows as Administrator on the Worker PC and run:

- CMD Window 1: `hdfs datanode`
- CMD Window 2: `yarn nodemanager`

Step 4: Verify Your Cluster

Go to your Master PC and open a web browser.

- Navigate to `http://172.16.0.81:9870` (HDFS Dashboard). Click on the "Datanodes" tab. You should see `172.16.0.82` listed as an active node!
- Navigate to `http://172.16.0.81:8088` (YARN Dashboard). Check the "Nodes" section to verify your worker is ready to accept MapReduce jobs.